

Cross-Modal Agreement-Aware Uncertainty-Calibrated Video-IMU Latent World Model

Executive summary: This project proposes a task-specific latent world model that learns how human gait evolves over time from synchronized video and wearable inertial measurements. The model will use current observations to forecast near-future freezing-of-gait (FOG) events or operational motor phases, while also estimating whether its forecast should be trusted.

The central research idea is agreement-aware forecasting. A video branch and an IMU branch will independently predict the same future target at the same horizon. Their agreement will be measured before fusion. If the branches disagree, the system will treat that disagreement as evidence that the forecast may be unreliable. A probabilistic latent dynamics model will then roll the fused state forward and decode possible future FOG outcomes.

The primary dataset is the public multimodal dataset of video, acceleration, angular velocity, and clinical scales collected during a turning-in-place task in 35 people with Parkinson's disease. Movement-disorder specialists reviewed the videos and identified freezing events, which were then mapped to the IMU timeline [1, 2].

Bottom line The project is not a clinical diagnosis system, a general-purpose physical simulator, or a robot-control system. It is a reliable future-event forecasting pipeline: video + IMU -> latent gait state -> future FOG prediction -> calibrated uncertainty.

1. Project definition and research logic

1.1 Problem being addressed

Many FOG systems are reactive: they examine a current sensor window and classify what is happening now. That is useful, but it does not explicitly model how the person's movement state is evolving into the near future, and it does not guarantee that a high confidence score is trustworthy. This project addresses both gaps by combining future-state forecasting with calibrated uncertainty.

1.2 Main research question

Research question Can synchronized video and wearable-IMU models forecast the same near-future gait state, and can their agreement or disagreement provide a calibrated signal of forecast reliability under ambiguous, missing, or unfamiliar conditions?

1.3 Project objectives

1. Build a subject-disjoint video-IMU forecasting benchmark from the public FOG dataset.
2. Fine-tune established video and time-series models to forecast the same future motor target.
3. Measure cross-modal agreement before fusion and test whether disagreement predicts forecast unreliability.
4. Use a probabilistic latent dynamics model to produce stochastic multi-step future rollouts.
5. Evaluate prediction quality, early warning, calibration, missing-modality behavior, and robustness.

1.4 Scope boundaries

Scope item	Decision
Included	Video + acceleration + angular velocity; future FOG/motor-phase forecasting; latent state; stochastic rollout; calibrated uncertainty; subject-disjoint evaluation.
Not included	Clinical diagnosis; a new backbone architecture; general-purpose physical simulation; deployed robot control; raw future RGB-video generation.
Primary claim	A task-specific multimodal latent world model can forecast near-future FOG-related states and provide a more honest reliability signal than a single modality alone.
Future extension	Conditional latent diffusion for sampling future latent, pose, or IMU trajectories after the core FYDP is complete.

2. Dataset and benchmark analysis

2.1 Recommended primary dataset

The recommended primary data source is the public dataset record titled "A public dataset of video, acceleration, angular velocity, and clinical scales in individuals with Parkinson's disease during the turning-in-place task." It contains data from 35 people with Parkinson's disease. During the task, participants alternated 360-degree turns to the right and left for approximately two minutes

at a self-selected pace. The dataset includes video trials, processed IMU files, metadata, and specialist-reviewed freezing events [1, 2].

Dataset record: [Figshare 14984667](#)

2.2 Why it fits this project

Requirement	Dataset fit
Video availability	Provides video trials that can be processed by a visual encoder.
IMU availability	Provides acceleration and angular-velocity signals for the same trials.
Event supervision	Specialists reviewed videos and identified freezing events, which were mapped to the IMU timeline.
Multimodal alignment	The same trial contains both visual and inertial evidence, enabling sample-level fusion after timestamp audit.
Research role	Primary multimodal benchmark/testbed for video-only, IMU-only, naive-fusion, and agreement-aware models.

2.3 Label design for the proposed task

The dataset's specialist freezing annotations should be converted into operational forecasting labels rather than interpreted as longitudinal clinical disease stages. A practical initial label design is: normal movement, pre-onset window, FOG episode, and recovery. The pre-onset label is important because the project aims to forecast an event before or around its onset, not simply detect it after it begins.

Important label distinction StartHesitation, Turn, and Walking are labels associated with the tdcsfog/defog sensor datasets. They

should not be assumed to be the native label structure of the video-IMU Figshare dataset. For this multimodal experiment, the target should be defined from the specialist freezing-event boundaries and documented before training.

2.4 Benchmark split and leakage control

- Use participant-level group splits; no participant should appear in both training and test sets.
- Do not allow overlapping windows from the same trial to cross a split boundary.
- Fit normalization, calibration, and threshold-selection procedures on training/validation data only.
- Keep a final subject-disjoint test set untouched until the analysis is frozen.
- Audit video-to-IMU file mapping and timestamps before constructing any training sample.

2.5 Dataset limitations

- The participant count is modest, so subject-level generalization is more important than the number of sliding windows.
- The turning-in-place task is controlled and may not represent ordinary walking or home-based behavior.
- Video and IMU timestamps, sampling rates, file naming, and annotation resolution must be audited before choosing final horizons.
- A sensor-only dataset such as tdcsfog/defog cannot by itself validate sample-level video-IMU fusion; it can only support an

optional sensor-domain robustness experiment unless aligned video is available.

3. Proposed system architecture

The architecture is deliberately built from established components. The research contribution is the way the components are trained, compared, calibrated, and evaluated, not the invention of a new backbone.

3.1 End-to-end workflow

1. Aligned observations	2. Encoders	3. Same-target forecasts	4. Agreement check	5. World-model rollout
Video frames + acceleration + angular velocity from the same time window.	DINOv2/visual encoder for frames; PatchTST or equivalent time-series encoder for IMU.	Video and IMU branches independently predict the same future phase at the same horizon.	Compare calibrated predictive distributions using Jensen-Shannon divergence.	Fuse context, roll latent state forward stochastically, and decode future FOG probabilities plus uncertainty.

Recommended initial setup: a 5-second observation context and a 2-second primary prediction horizon, with shorter-horizon sensitivity checks after confirming the dataset's sampling and annotation resolution.

3.2 Component roles

Component	Role
Visual encoder	DINOv2 ViT-S/14, initially frozen or lightly fine-tuned; extracts visual features from frames.
Video temporal model	A standard Transformer encoder aggregates frame features over the observation window.
IMU model	PatchTST or another established time-series model represents acceleration and angular velocity.
Fusion	Standard late probability fusion combines calibrated modality predictions; no custom fusion backbone is required.
World model	Dreamer-style RSSM learns probabilistic latent transitions and multi-step rollouts.
Uncertainty	Cross-modal disagreement, ensemble variance, predictive entropy, and transition variance are combined and calibrated.

3.3 Why this qualifies as a world model

- It maintains a compact latent state representing the current gait context.
- It learns probabilistic transitions from the current latent state toward future latent states.
- It produces multi-step stochastic rollouts rather than only a single reactive label.
- It decodes the rolled-out states into future motor or FOG-related predictions.

Evaluation principle Latent rollout consistency is a useful diagnostic, but the primary performance measure must be decoded future-event or future-phase accuracy because the learned latent state has no independent ground-truth label.

4. Training and experimental design

4.1 Data preparation

- Inspect the dataset archive, file naming, metadata, trial IDs, video frame rates, IMU sampling rates, and annotation files.
- Create a deterministic video-to-IMU timestamp map and document any resampling or synchronization correction.
- Construct fixed observation windows and future target windows without crossing trial boundaries.
- Generate operational labels from specialist freezing-event boundaries and freeze the label protocol before model training.
- Apply training-only normalization and preserve the same transformation for validation and test samples.

4.2 Recommended experiment ladder

Experiment	Purpose
A. Video-only	Tests how much visual evidence can predict the future target.
B. IMU-only	Tests how much inertial evidence can predict the future target.
C. Naive fusion	Combines modalities without using agreement as an explicit reliability signal.

Experiment	Purpose
D. Agreement-aware fusion	Measures pre-fusion agreement/disagreement and uses it in uncertainty estimation.
E. Probabilistic latent world model	Adds stochastic latent transitions and multi-step rollouts.
F. Missing-modality tests	Removes or corrupts video or IMU to test whether uncertainty responds appropriately.

4.3 Uncertainty design

Source	Operational measure
Cross-modal disagreement	Jensen-Shannon divergence between calibrated video and IMU predictive distributions.
Aleatoric uncertainty	Predictive entropy or heteroscedastic transition variance; uncertainty caused by intrinsic ambiguity.
Epistemic uncertainty	Variance across 3-5 independently trained model seeds; uncertainty caused by limited knowledge.
Calibration	Temperature scaling on validation data, then ECE, Brier score, NLL, reliability diagrams, and coverage/risk analysis.
Calibration rule Calibration must be learned on	

Source	Operational measure
validation data and frozen before final testing. A high confidence score is useful only when its empirical correctness is consistent with the stated confidence.	

5. Evaluation framework

The project should evaluate prediction quality and trustworthiness together. A model that predicts well but is confidently wrong under uncertainty is not sufficient for the stated research goal.

Dimension	Measures	What it answers
Future-event prediction	AUPRC, macro F1, balanced accuracy, decoded future-state accuracy	Whether the model forecasts the future target correctly.
Early warning	Event-onset lead time	Whether the model gives useful warning before the annotated event begins.

Dimension	Measures	What it answers
Probabilistic quality	NLL and Brier score	Whether predicted distributions assign probability appropriately.
Calibration	ECE, reliability diagrams, calibration gap	Whether stated confidence matches observed correctness.
Selective reliability	Risk-coverage curves and coverage analysis	Whether the model can abstain or defer when it is uncertain.
Robustness	Missing-modality tests, corruption tests, optional external sensor shift	Whether uncertainty rises when evidence is incomplete or unfamiliar.

5.1 Key comparisons

- Video-only versus IMU-only: establishes the contribution of each modality.
- Naive fusion versus agreement-aware fusion: tests whether explicit disagreement improves reliability estimation.
- Deterministic versus probabilistic latent dynamics: tests the value of stochastic rollouts.
- Familiar versus corrupted or missing-modality inputs: tests whether uncertainty behaves sensibly.
- Short versus longer horizons: tests whether uncertainty increases as the future becomes harder to determine.

5.2 Evidence needed for the main claim

The strongest evidence would be a consistent pattern in which agreement-aware fusion maintains or improves future-event performance while producing lower calibration error, better reliability diagrams, and higher uncertainty when one modality is corrupted, missing, or unfamiliar. A simple accuracy gain without improved reliability would not fully support the project claim.

External validation caution The primary Figshare dataset supports sample-level video-IMU fusion. The tdcsfog/defog and FoG-STAR datasets can support optional sensor-only robustness or transfer analyses, but they should not be presented as video-IMU fusion validation unless synchronized video is available for the same samples.

6. Expected contribution and feasibility

6.1 Expected scientific contribution

- A reproducible multimodal forecasting pipeline for near-future FOG-related motor states.
- A same-target pre-fusion agreement test that turns modality disagreement into a reliability signal.
- A task-specific probabilistic latent world model with decoded multi-step forecasts.
- An evaluation protocol that reports prediction quality, calibration, missing modality, and uncertainty under change together.

6.2 What is novel and what is not

Question	Analysis
Not the main novelty	Inventing a new visual backbone, a new time-series backbone, or a generic FOG classifier.
Potential novelty	The intersection of same-target multi-horizon video-IMU forecasting, cross-modal agreement-aware uncertainty, probabilistic latent rollout, and leakage-free evaluation.
Defensibility condition	The paper must show that disagreement and uncertainty are empirically useful, not only add them as decorative scores.

6.3 Compute and implementation feasibility

Resource	Recommendation
GPU	One RTX 3060 12 GB is workable if DINOv2 is frozen and visual embeddings are cached; 16 GB is more comfortable.
Memory/storage	Approximately 32 GB RAM and 150-300 GB SSD are recommended for cached features, intermediate windows, and repeated runs.
Model policy	Use established components and keep the backbone frozen or lightly fine-tuned initially.
Out of scope	Full V-JEPA-style pretraining and raw RGB video diffusion are too large for the core FYDP.

6.4 Indicative implementation timeline

Period	Milestone
Weeks 1-2	Literature review, data/license audit, and final protocol.
Weeks 3-5	Timestamp alignment, label construction, and subject-disjoint split.
Weeks 6-8	Video-only and IMU-only baselines.
Weeks 9-11	Same-target multi-horizon forecasting and calibration.
Weeks 12-15	RSSM-based probabilistic latent rollout.
Weeks 16-18	Deep ensemble, missing modality, and robustness tests.
Weeks 19-24	Ablations, statistical analysis, writing, and reproducible code release.

7. Risks, controls, and decisions required

Risk	Why it matters	Control
Small participant count	The model may overfit windows from a small number of people.	Use participant-level splits, conservative fine-tuning, augmentation, and report subject-level uncertainty.

Risk	Why it matters	Control
Timestamp mismatch	Video and IMU may not align exactly.	Audit trial IDs and timestamps; document resampling and test sensitivity to alignment.
Leakage from windows	Overlapping windows can make test performance look artificially strong.	Split by participant and trial before window generation; never split windows randomly.
Sparse FOG events	Accuracy can look high even when rare events are missed.	Use AUPRC, event-onset lead time, class-weighted loss, and event-level analysis.
Unclear label boundary	Different onset rules can change the target.	Pre-register an operational label protocol and run a small annotation-resolution sensitivity analysis.
Compute cost	Raw video processing can exceed the FYDP budget.	Freeze/cache DINOv2 features and train temporal models on cached embeddings.
Overclaiming domain shift	The main dataset is a controlled turning task.	Describe external sensor datasets as optional robustness evidence, not as full video-domain validation.

7.1 Decisions to freeze before coding

- Final event/phase labels and onset definition.

- Primary observation window and prediction horizons.
- Participant-level split and final held-out test participants.
- Exact video preprocessing and IMU synchronization procedure.
- Which metadata or clinical scales, if any, are allowed as conditioning variables.
- The calibration method and the threshold-selection rule for event onset.
- Whether external sensor-only datasets are included as optional robustness experiments.

Go/no-go recommendation Proceed with the video-IMU project if the dataset audit confirms reliable trial-level synchronization and enough usable freezing events for participant-disjoint evaluation. Keep the contribution focused on forecasting reliability and uncertainty; do not expand the core scope to custom architectures or raw video generation.

8. Conclusion

The proposed study is a coherent and feasible multimodal world-model project when defined as a task-specific forecast of near-future freezing-of-gait states. The public video-IMU dataset is a suitable benchmark/testbed because it contains synchronized visual and inertial evidence with specialist-reviewed freezing events. The strongest research value lies in asking a model not only "what will happen next?" but also "how much should that forecast be trusted?"

The project should therefore be implemented as a controlled sequence: audit and align the dataset, define labels, build video-only

and IMU-only baselines, measure same-target agreement, add probabilistic latent rollout, calibrate uncertainty, and test behavior under missing or corrupted evidence. If the model becomes less confident when the evidence becomes less reliable, while maintaining useful future-event performance, the central research argument will be well supported.

Appendix A. Supervisor-ready project statement

One-paragraph version This project develops a task-specific, uncertainty-calibrated video-IMU latent world model for forecasting near-future freezing-of-gait events in Parkinson's disease. Using a public dataset containing synchronized video, acceleration, angular velocity, and specialist-reviewed freezing annotations, separate visual and inertial branches will forecast the same future motor phase at multiple horizons. Their predictive agreement will be measured before fusion, and disagreement will contribute to a calibrated reliability estimate. A probabilistic RSSM-style latent dynamics model will then generate stochastic multi-step rollouts and decode them into future FOG probabilities. The study will compare video-only, IMU-only, naive-fusion, and agreement-aware models using subject-disjoint splits, event-onset lead time, AUPRC, NLL, Brier score, ECE, reliability diagrams, and missing-modality tests.

Appendix B. Simple terminology

Term	Meaning
Latent state	A compact hidden summary of the person's current movement condition.
World model	A model that represents a current state, learns how it changes, and simulates possible future states.
Forecasting	Predicting what may happen in the next time window, rather than only classifying the current window.
Cross-modal agreement	How similarly the video and IMU branches predict the same future target.
Uncertainty	How unsure the model is about its forecast.
Calibration	Whether a confidence value matches the model's actual correctness over many cases.
Domain shift	A change in people, setting, task, sensor, or recording conditions that makes the test data different from training data.

References

- [1] Ribeiro de Souza, C. et al. (2022). A Public Data Set of Videos, Inertial Measurement Unit, and Clinical Scales of Freezing of Gait in Individuals With Parkinson's Disease During a Turning-In-Place Task. *Frontiers in Neuroscience*, 16, 832463. [Frontiers paper](#)
- [2] Figshare dataset record: A public dataset of video, acceleration, angular velocity, and clinical scales in individuals with Parkinson's

disease during the turning-in-place task. DOI:
10.6084/m9.figshare.14984667. [Figshare dataset record](#)

[3] Hafner, D. et al. (2023). Mastering Diverse Domains through World Models. arXiv:2301.04104. [World Models paper](#)

[4] Oquab, M. et al. (2023). DINOv2: Learning Robust Visual Features without Supervision. arXiv:2304.07193. [DINOv2 paper](#)

[5] Nie, Y. et al. (2023). A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. arXiv:2211.14730. [PatchTST paper](#)

Report basis: the three supplied proposal/concept documents and the public dataset record above. Where the supplied drafts differ, this report treats the video-IMU multimodal scope as the requested primary project and identifies accelerometer-only datasets as optional comparison or robustness resources.